KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5243A | 5 Credits: 4 | Program Elective | Source-grounded

Analytical Instruments

 To impart knowledge on the fundamentals of Analytical instruments

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5243A is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5243A.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Biomedical Engineering
Course code	5243A
Course title	Analytical Instruments
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

11 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart knowledge on the fundamentals of Analytical instruments
- To equip students to handle various analytical instruments which are useful for
- clinical analysis in hospitals and pharmaceutical laboratories.

Course outcomes

CO	Official outcome	Study level
CO1	Describe the fundamentals of Analytical Instrument. 14 Understanding Explain UV, Visible, IR Spectrophotometers and	See official syllabus
CO2	16 Understanding flame photometer Illustrate the instruments for Chromatography,	See official syllabus
CO3	14 Understanding Electrophoresis and densitometry. Explain pH meter, blood cell counters and blood	See official syllabus
CO4	14 Understanding Gas Analyzer	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Describe the fundamentals of Analytical Instruments.	3	As specified
Module 2	Explain UV, Visible, IR Spectrophotometers and flame photometer	2	As specified
Module 3	densitometry.	3	As specified
Module 4	Explain pH meter, blood cell counters and blood Gas Analyzer	3	As specified

MODULE 1

Describe the fundamentals of Analytical Instruments.

This module is presented from the official syllabus scope for Analytical Instruments. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe the fundamentals of Analytical Instruments.

- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

7 and its performance requirement Summarize the Instrument Calibration

7 and its performance requirement Summarize the Instrument Calibration is an official topic in Module 1 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 7 and its performance requirement Summarize the Instrument Calibration using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 7 and its performance requirement summarize the instrument calibration should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 7 and its performance requirement Summarize the Instrument Calibration means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-[illegible] 7 and its performance requirement Summarize the Instrument Calibration [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding Techniques & Validation

3 Understanding Techniques & Validation is an official topic in Module 1 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Techniques & Validation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding techniques & validation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Techniques & Validation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 3 Understanding Techniques & Validation [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain Beer-Lambert’s law and colorimeter 4 Understanding Fundamentals of Analytical Instruments Introduction to Analytical Instrumentation: Fundamentals of analytical instruments: Elements of an analytical instrument - PC based analytical instruments - Classification of instrumental techniques. Performance requirement-definition of accuracy, precision, sensitivity, resolution and error, need for calibration Analytical instruments used in clinical environment (listing) and its applications. Electromagnetic radiation - Electromagnetic spectrum- Laws relating to absorption of radiation. Beer-Lambert's Law. Colorimeters - block diagram, Principle &applications, Calibration procedure.

Explain Beer-Lambert’s law and colorimeter 4 Understanding Fundamentals of Analytical Instruments Introduction to Analytical Instrumentation: Fundamentals of analytical instruments: Elements of an analytical instrument - PC based analytical instruments - Classification of instrumental techniques. Performance requirement-definition of accuracy, precision, sensitivity, resolution and error, need for calibration Analytical instruments used in clinical environment (listing) and its applications. Electromagnetic radiation - Electromagnetic spectrum- Laws relating to absorption of radiation. Beer-Lambert's Law. Colorimeters - block diagram, Principle &applications, Calibration procedure. is an official topic in Module 1 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Beer-Lambert’s law and colorimeter 4 Understanding Fundamentals of Analytical Instruments Introduction to Analytical Instrumentation: Fundamentals of analytical instruments: Elements of an analytical instrument - PC based analytical instruments - Classification of instrumental techniques. Performance requirement-definition of accuracy, precision, sensitivity, resolution and error, need for calibration Analytical instruments used in clinical environment (listing) and its applications. Electromagnetic radiation - Electromagnetic spectrum- Laws relating to absorption of radiation. Beer-Lambert's Law. Colorimeters - block diagram, Principle &applications, Calibration procedure. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain beer-lambert’s law and colorimeter 4 understanding fundamentals of analytical instruments introduction to analytical instrumentation: fundamentals of analytical instruments: elements of an analytical instrument - pc based analytical instruments - classification of instrumental techniques. performance requirement-definition of accuracy, precision, sensitivity, resolution and error, need for calibration analytical instruments used in clinical environment (listing) and its applications. electromagnetic radiation - electromagnetic spectrum- laws relating to absorption of radiation. beer-lambert's law. colorimeters - block diagram, principle &applications, calibration procedure. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Beer-Lambert’s law and colorimeter 4 Understanding Fundamentals of Analytical Instruments Introduction to Analytical Instrumentation: Fundamentals of analytical instruments: Elements of an analytical instrument - PC based analytical instruments - Classification of instrumental techniques. Performance requirement-definition of accuracy, precision, sensitivity, resolution and error, need for calibration Analytical instruments used in clinical environment (listing) and its applications. Electromagnetic radiation - Electromagnetic spectrum- Laws relating to absorption of radiation. Beer-Lambert's Law. Colorimeters - block diagram, Principle &applications, Calibration procedure. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain Beer-Lambert’s law and colorimeter 4 Understanding Fundamentals of Analytical Instruments Introduction to Analytical Instrumentation: Fundamentals of analytical instruments: Elements of an analytical instrument - PC based analytical instruments - Classification of instrumental techniques. Performance requirement-definition of accuracy, precision, sensitivity, resolution and error, need for calibration Analytical instruments used in clinical environment (listing) and its applications. Electromagnetic radiation - Electromagnetic spectrum- Laws relating to absorption of radiation. Beer-Lambert's Law. Colorimeters - block diagram, Principle &applications, Calibration procedure. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Describe the fundamentals of Analytical Instruments.

            Describe the elements of Analytical Instrument
Understanding
  M1.01                                                         7
            and its performance requirement
            Summarize the Instrument Calibration
  M1.02                                                         3
Understanding
            Techniques & Validation
  M1.03     Explain Beer-Lambert’s law and colorimeter              4
Understanding

Contents:
Fundamentals of Analytical Instruments
Introduction to Analytical Instrumentation: Fundamentals of analytical
instruments:
Elements of an analytical instrument - PC based analytical instruments -
Classification of
instrumental techniques.
Performance requirement-definition of accuracy, precision, sensitivity,
resolution and error,
need for calibration
Analytical instruments used in clinical environment (listing) and its
applications.
Electromagnetic radiation - Electromagnetic spectrum- Laws relating to absorption
of
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Explain UV, Visible, IR Spectrophotometers and flame photometer

This module is presented from the official syllabus scope for Analytical Instruments. Use the topic cards for learning and the source transcript for exact coverage.

As specified	CO-linked
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain UV, Visible, IR Spectrophotometers and flame photometer
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

9 Understanding and IR spectrophotometers Illustrate the working principle and

9 Understanding and IR spectrophotometers Illustrate the working principle and is an official topic in Module 2 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 9 Understanding and IR spectrophotometers Illustrate the working principle and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 9 understanding and ir spectrophotometers illustrate the working principle and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 9 Understanding and IR spectrophotometers Illustrate the working principle and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 9 Understanding and IR spectrophotometers Illustrate the working principle and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding constructional details of flame photometer. SPECTROPHOTOMETER Principle of absorption spectroscopy- Types(Listing) Spectrophotometers: basic sub components - monochromator, detection systems and amplifiers - single beam & double beam - visible, UV & IR (basics only), - applications in biochemical analysis. Flame Photometry: Principle and constructional details of flame photometer- Emission system - Optical system - Detectors. Applications of flame photometry. Illustrate the instruments for Chromatography, Electrophoresis and

2 Understanding constructional details of flame photometer. SPECTROPHOTOMETER Principle of absorption spectroscopy- Types(Listing) Spectrophotometers: basic sub components - monochromator, detection systems and amplifiers - single beam & double beam - visible, UV & IR (basics only), - applications in biochemical analysis. Flame Photometry: Principle and constructional details of flame photometer- Emission system - Optical system - Detectors. Applications of flame photometry. Illustrate the instruments for Chromatography, Electrophoresis and is an official topic in Module 2 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding constructional details of flame photometer. SPECTROPHOTOMETER Principle of absorption spectroscopy- Types(Listing) Spectrophotometers: basic sub components - monochromator, detection systems and amplifiers - single beam & double beam - visible, UV & IR (basics only), - applications in biochemical analysis. Flame Photometry: Principle and constructional details of flame photometer- Emission system - Optical system - Detectors. Applications of flame photometry. Illustrate the instruments for Chromatography, Electrophoresis and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding constructional details of flame photometer. spectrophotometer principle of absorption spectroscopy- types(listing) spectrophotometers: basic sub components - monochromator, detection systems and amplifiers - single beam & double beam - visible, uv & ir (basics only), - applications in biochemical analysis. flame photometry: principle and constructional details of flame photometer- emission system - optical system - detectors. applications of flame photometry. illustrate the instruments for chromatography, electrophoresis and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding constructional details of flame photometer. SPECTROPHOTOMETER Principle of absorption spectroscopy- Types(Listing) Spectrophotometers: basic sub components - monochromator, detection systems and amplifiers - single beam & double beam - visible, UV & IR (basics only), - applications in biochemical analysis. Flame Photometry: Principle and constructional details of flame photometer- Emission system - Optical system - Detectors. Applications of flame photometry. Illustrate the instruments for Chromatography, Electrophoresis and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Understanding constructional details of flame photometer. SPECTROPHOTOMETER Principle of absorption spectroscopy- Types(Listing) Spectrophotometers: basic sub components - monochromator, detection systems and amplifiers - single beam & double beam - visible, UV & IR (basics only), - applications in biochemical analysis. Flame Photometry: Principle and constructional details of flame photometer- Emission system - Optical system - Detectors. Applications of flame photometry. Illustrate the instruments for Chromatography, Electrophoresis and □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain UV, Visible, IR Spectrophotometers and flame photometer

            Explain basic components of
  M1.01                                                           5
Understanding
            spectrophotometer
            Describe the working principle of UV, Visible
  M2.01                                                           9
Understanding
            and IR spectrophotometers
            Illustrate the working principle and
  M2.02                                                           2
Understanding
            constructional details of flame photometer.
            Series Test – I                                       1

Contents:
SPECTROPHOTOMETER
Principle of absorption spectroscopy- Types(Listing)
Spectrophotometers: basic sub components - monochromator, detection systems and
amplifiers - single beam & double beam - visible, UV & IR (basics only), -
applications in
biochemical analysis.
Flame Photometry: Principle and constructional details of flame photometer-
Emission
system - Optical system - Detectors. Applications of flame photometry.
            Illustrate the instruments for Chromatography, Electrophoresis and
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

densitometry.

This module is presented from the official syllabus scope for Analytical Instruments. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: densitometry.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain chromatography and its types. 3 Understanding Explain the principle and construction details of

Explain chromatography and its types. 3 Understanding Explain the principle and construction details of is an official topic in Module 3 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain chromatography and its types. 3 Understanding Explain the principle and construction details of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain chromatography and its types. 3 understanding explain the principle and construction details of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain chromatography and its types. 3 Understanding Explain the principle and construction details of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Explain chromatography and its types. 3 Understanding Explain the principle and construction details of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

6 gas and liquid chromatography in detail. Understanding Describe the working of electrophoresis machine

6 gas and liquid chromatography in detail. Understanding Describe the working of electrophoresis machine is an official topic in Module 3 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 gas and liquid chromatography in detail. Understanding Describe the working of electrophoresis machine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 gas and liquid chromatography in detail. understanding describe the working of electrophoresis machine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 gas and liquid chromatography in detail. Understanding Describe the working of electrophoresis machine means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 6 gas and liquid chromatography in detail. Understanding Describe the working of electrophoresis machine □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

5 Understanding and densitometer Chromatography And Electrophoresis Chromatographic process - Classification- Terms in chromatography - Gas chromatography: Block diagram - Principle - Constructional details Liquid Chromatography: Types of liquid chromatography- High pressure Liquid Chromatography (HPLC): Principle - Constructional details, Applications in medical field. Electrophoresis apparatus - basic principle and block diagram. Densitometer - principle and block diagram

5 Understanding and densitometer Chromatography And Electrophoresis Chromatographic process - Classification- Terms in chromatography - Gas chromatography: Block diagram - Principle - Constructional details Liquid Chromatography: Types of liquid chromatography- High pressure Liquid Chromatography (HPLC): Principle - Constructional details, Applications in medical field. Electrophoresis apparatus - basic principle and block diagram. Densitometer - principle and block diagram is an official topic in Module 3 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding and densitometer Chromatography And Electrophoresis Chromatographic process - Classification- Terms in chromatography - Gas chromatography: Block diagram - Principle - Constructional details Liquid Chromatography: Types of liquid chromatography- High pressure Liquid Chromatography (HPLC): Principle - Constructional details, Applications in medical field. Electrophoresis apparatus - basic principle and block diagram. Densitometer - principle and block diagram using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding and densitometer chromatography and electrophoresis chromatographic process - classification- terms in chromatography - gas chromatography: block diagram - principle - constructional details liquid chromatography: types of liquid chromatography- high pressure liquid chromatography (hplc): principle - constructional details, applications in medical field. electrophoresis apparatus - basic principle and block diagram. densitometer - principle and block diagram should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding and densitometer Chromatography And Electrophoresis Chromatographic process - Classification- Terms in chromatography - Gas chromatography: Block diagram - Principle - Constructional details Liquid Chromatography: Types of liquid chromatography- High pressure Liquid Chromatography (HPLC): Principle - Constructional details, Applications in medical field. Electrophoresis apparatus - basic principle and block diagram. Densitometer - principle and block diagram means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 5 Understanding and densitometer Chromatography And Electrophoresis Chromatographic process - Classification- Terms in chromatography - Gas chromatography: Block diagram - Principle - Constructional details Liquid Chromatography: Types of liquid chromatography- High pressure Liquid Chromatography (HPLC): Principle - Constructional details, Applications in medical field. Electrophoresis apparatus - basic principle and block diagram. Densitometer - principle and block diagram □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
densitometry.

  M3.01     Explain chromatography and its types.                  3

Understanding
            Explain the principle and construction details of
  M3.02                                                            6
            gas and liquid chromatography in detail.
Understanding
            Describe the working of electrophoresis machine
  M3.03                                                            5
Understanding
            and densitometer

Contents:
Chromatography And Electrophoresis
Chromatographic process - Classification- Terms in chromatography - Gas
chromatography:
Block diagram - Principle - Constructional details
Liquid Chromatography: Types of liquid chromatography- High pressure Liquid
Chromatography (HPLC): Principle - Constructional details, Applications in medical
field.
Electrophoresis apparatus - basic principle and block diagram. Densitometer -
principle and
block diagram
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Explain pH meter, blood cell counters and blood Gas Analyzer

This module is presented from the official syllabus scope for Analytical Instruments. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain pH meter, blood cell counters and blood Gas Analyzer
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

3 Understanding meter. Detail the working of blood cell counters and

3 Understanding meter. Detail the working of blood cell counters and is an official topic in Module 4 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding meter. Detail the working of blood cell counters and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding meter. detail the working of blood cell counters and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding meter. Detail the working of blood cell counters and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-[illegible] 3 Understanding meter. Detail the working of blood cell counters and [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

6 Understanding Hb meter. Describe the principle & construction of

6 Understanding Hb meter. Describe the principle & construction of is an official topic in Module 4 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding Hb meter. Describe the principle & construction of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding hb meter. describe the principle & construction of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding Hb meter. Describe the principle & construction of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-□□ 6 Understanding Hb meter. Describe the principle & construction of □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

5 Understanding Blood Gas Analyzer and glucometer pH METER, BLOOD CELL COUNTERS & BLOOD GAS ANALYZER pH meter - Principle of pH measurement - Electrodes for pH measurement - Block diagram of pH meter. Types of blood cells, Blood cell counters - types - coulter counter and flow cytometry. Auto analyzers - block diagram and working. Hb meter - block diagram Blood Gas Analyzer - pO2 and pCO2 measurement - Principle, construction and working of PO2 electrode and PCO2 electrode - Blood Gas Analyzer - block diagram of blood gas analyser. Glucometer - working principle and basic block diagram Text / Reference: T/R Book Title/Author R.S. Khandpur, 'Handbook of Analytical Instruments', Tata McGraw Hill T1 publishing Co. Ltd., 2003. T2 G.W. Ewing, Instrumental Methods of Analysis, Mc Graw Hill, 2004. R1 Robrt D Braun' , 'Introduction to Instrumental Analysis' , Mc Graw Hill Ltd Galen W Ewing, 'Instrumental methods of chemical analysis' , Mc Graw Hill R2 Ltd Online Resources: Sl.No Website Link 1 https://www.hemakumar.in/analytical-instrumentation 2 https://www.slideshare.net/ShreyaAhuja2/ph-meter-54084084 3 https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html#

5 Understanding Blood Gas Analyzer and glucometer pH METER, BLOOD CELL COUNTERS & BLOOD GAS ANALYZER pH meter - Principle of pH measurement - Electrodes for pH measurement - Block diagram of pH meter. Types of blood cells, Blood cell counters - types - coulter counter and flow cytometry. Auto analyzers - block diagram and working. Hb meter - block diagram Blood Gas Analyzer - pO2 and pCO2 measurement - Principle, construction and working of PO2 electrode and PCO2 electrode - Blood Gas Analyzer - block diagram of blood gas analyser. Glucometer - working principle and basic block diagram Text / Reference: T/R Book Title/Author R.S. Khandpur, 'Handbook of Analytical Instruments', Tata McGraw Hill T1 publishing Co. Ltd., 2003. T2 G.W. Ewing, Instrumental Methods of Analysis, Mc Graw Hill, 2004. R1 Robrt D Braun' , 'Introduction to Instrumental Analysis' , Mc Graw Hill Ltd Galen W Ewing, 'Instrumental methods of chemical analysis' , Mc Graw Hill R2 Ltd Online Resources: Sl.No Website Link 1 https://www.hemakumar.in/analytical-instrumentation 2 https://www.slideshare.net/ShreyaAhuja2/ph-meter-54084084 3 https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html# is an official topic in Module 4 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding Blood Gas Analyzer and glucometer pH METER, BLOOD CELL COUNTERS & BLOOD GAS ANALYZER pH meter - Principle of pH measurement - Electrodes for pH measurement - Block diagram of pH meter. Types of blood cells, Blood cell counters - types - coulter counter and flow cytometry. Auto analyzers - block diagram and working. Hb meter - block diagram Blood Gas Analyzer - pO2 and pCO2 measurement - Principle, construction and working of PO2 electrode and PCO2 electrode - Blood Gas Analyzer - block diagram of blood gas analyser. Glucometer - working principle and basic block diagram Text / Reference: T/R Book Title/Author R.S. Khandpur, ‘Handbook of Analytical Instruments’, Tata McGraw Hill T1 publishing Co. Ltd., 2003. T2 G.W. Ewing, Instrumental Methods of Analysis, Mc Graw Hill, 2004. R1 Robrt D Braun’ , ‘Introduction to Instrumental Analysis’ , Mc Graw Hill Ltd Galen W Ewing, ‘Instrumental methods of chemical analysis’ , Mc Graw Hill R2 Ltd Online Resources: Sl.No Website Link 1 https://www.hemakumar.in/analytical-instrumentation 2 https://www.slideshare.net/ShreyaAhuja2/ph-meter-54084084 3 https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html# using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding blood gas analyzer and glucometer ph meter, blood cell counters & blood gas analyzer ph meter - principle of ph measurement - electrodes for ph measurement - block diagram of ph meter. types of blood cells, blood cell counters - types - coulter counter and flow cytometry. auto analyzers - block diagram and working. hb meter - block diagram blood gas analyzer - po2 and pco2 measurement - principle, construction and working of po2 electrode and pco2 electrode - blood gas analyzer - block diagram of blood gas analyser. glucometer - working principle and basic block diagram text / reference: t/r book title/author r.s. khandpur, ‘handbook of analytical instruments’, tata mcgraw hill t1 publishing co. ltd., 2003. t2 g.w. ewing, instrumental methods of analysis, mc graw hill, 2004. r1 robrt d braun’ , ‘introduction to instrumental analysis’ , mc graw hill ltd galen w ewing, ‘instrumental methods of chemical analysis’ , mc graw hill r2 ltd online resources: sl.no website link 1 https://www.hemakumar.in/analytical-instrumentation 2 https://www.slideshare.net/shreyaahuja2/ph-meter-54084084 3 https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html# should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding Blood Gas Analyzer and glucometer pH METER, BLOOD CELL COUNTERS & BLOOD GAS ANALYZER pH meter - Principle of pH measurement - Electrodes for pH measurement - Block diagram of pH meter. Types of blood cells, Blood cell counters - types - coulter counter and flow cytometry. Auto analyzers - block diagram and working. Hb meter - block diagram Blood Gas Analyzer - pO2 and pCO2 measurement - Principle, construction and working of PO2 electrode and PCO2 electrode - Blood Gas Analyzer - block diagram of blood gas analyser. Glucometer - working principle and basic block diagram Text / Reference: T/R Book Title/Author R.S. Khandpur, ‘Handbook of Analytical Instruments’, Tata McGraw Hill T1 publishing Co. Ltd., 2003. T2 G.W. Ewing, Instrumental Methods of Analysis, Mc Graw Hill, 2004. R1 Robrt D Braun’ , ‘Introduction to Instrumental Analysis’ , Mc Graw Hill Ltd Galen W Ewing, ‘Instrumental methods of chemical analysis’ , Mc Graw Hill R2 Ltd Online Resources: Sl.No Website Link 1 https://www.hemakumar.in/analytical-instrumentation 2 https://www.slideshare.net/ShreyaAhuja2/ph-meter-54084084 3 https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html# means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 5 Understanding Blood Gas Analyzer and glucometer pH METER, BLOOD CELL COUNTERS & BLOOD GAS ANALYZER pH meter - Principle of pH measurement - Electrodes for pH measurement - Block diagram of pH meter. Types of blood cells, Blood cell counters - types - coulter counter and flow cytometry. Auto analyzers - block diagram and working. Hb meter - block diagram Blood Gas Analyzer - pO2 and pCO2 measurement - Principle, construction and working of PO2 electrode and PCO2 electrode - Blood Gas Analyzer - block diagram of blood gas analyser. Glucometer - working principle and basic block diagram Text / Reference: T/R Book Title/Author R.S. Khandpur, ‘Handbook of Analytical Instruments’, Tata McGraw Hill T1 publishing Co. Ltd., 2003. T2 G.W. Ewing, Instrumental Methods of Analysis, Mc Graw Hill, 2004. R1 Robrt D Braun’ , ‘Introduction to Instrumental Analysis’ , Mc Graw Hill Ltd Galen W Ewing, ‘Instrumental methods of chemical analysis’ , Mc Graw Hill R2 Ltd Online Resources: Sl.No Website Link 1 https://www.hemakumar.in/analytical-instrumentation 2 https://www.slideshare.net/ShreyaAhuja2/ph-meter-54084084 3 https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html# □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain pH meter, blood cell counters and blood Gas Analyzer
            Explain the principle & construction of pH
  M4.01                                                      3
Understanding
            meter.
            Detail the working of blood cell counters and
  M4.02                                                      6
Understanding
            Hb meter.
            Describe the principle & construction of
  M4.03                                                      5
Understanding
            Blood Gas Analyzer and glucometer
            Series Test – II                                 1

Contents:
pH METER, BLOOD CELL COUNTERS & BLOOD GAS ANALYZER
pH meter - Principle of pH measurement - Electrodes for pH measurement - Block
diagram
of pH meter.
Types of blood cells, Blood cell counters - types - coulter counter and flow
cytometry. Auto
analyzers - block diagram and working. Hb meter - block diagram
Blood Gas Analyzer - pO2 and pCO2 measurement - Principle, construction and
working of
PO2 electrode and PCO2 electrode - Blood Gas Analyzer - block diagram of blood
gas
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 7 and its performance requirement Summarize the Instrument Calibration. (3 marks)

Answer: Begin with the standard definition or identity of *7 and its performance requirement Summarize the Instrument Calibration*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding Techniques & Validation. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding Techniques & Validation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain Beer-Lambert’s law and colorimeter 4 Understanding Fundamentals of Analytical Instruments Introduction to Analytical Instrumentation: Fundamentals of analytical instruments: Elements of an analytical instrument - PC based analytical instruments - Classification of instrumental techniques. Performance requirement-definition of accuracy, precision, sensitivity, resolution and error, need for calibration Analytical instruments used in clinical environment (listing) and its applications. Electromagnetic radiation - Electromagnetic spectrum- Laws relating to absorption of radiation. Beer-Lambert's Law. Colorimeters - block diagram, Principle &applications, Calibration procedure.. (3 marks)

Answer: Begin with the standard definition or identity of *Explain Beer-Lambert’s law and colorimeter 4 Understanding Fundamentals of Analytical Instruments Introduction to Analytical Instrumentation: Fundamentals of analytical instruments: Elements of an analytical instrument - PC based analytical instruments - Classification of instrumental techniques. Performance requirement-definition of accuracy, precision, sensitivity, resolution and error, need for calibration Analytical instruments used in clinical environment (listing) and its applications. Electromagnetic radiation - Electromagnetic spectrum- Laws relating to absorption of radiation. Beer-Lambert's Law. Colorimeters - block diagram, Principle &applications, Calibration procedure.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 9 Understanding and IR spectrophotometers Illustrate the working principle and. (3 marks)

Answer: Begin with the standard definition or identity of *9 Understanding and IR spectrophotometers Illustrate the working principle and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding constructional details of flame photometer. SPECTROPHOTOMETER Principle of absorption spectroscopy- Types(Listing) Spectrophotometers: basic sub components - monochromator, detection systems and amplifiers - single beam & double beam - visible, UV & IR (basics only), - applications in biochemical analysis. Flame Photometry: Principle and constructional details of flame photometer- Emission system - Optical system - Detectors. Applications of flame photometry. Illustrate the instruments for Chromatography, Electrophoresis and. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding constructional details of flame photometer. SPECTROPHOTOMETER Principle of absorption spectroscopy- Types(Listing) Spectrophotometers: basic sub components - monochromator, detection systems and amplifiers - single beam & double beam - visible, UV & IR (basics only), - applications in biochemical analysis. Flame Photometry: Principle and constructional details of flame photometer- Emission system - Optical system - Detectors. Applications of flame photometry. Illustrate the instruments for Chromatography, Electrophoresis and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Explain chromatography and its types. 3 Understanding Explain the principle and construction details of. (3 marks)

Answer: Begin with the standard definition or identity of *Explain chromatography and its types. 3 Understanding Explain the principle and construction details of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 6 gas and liquid chromatography in detail. Understanding Describe the working of electrophoresis machine. (3 marks)

Answer: Begin with the standard definition or identity of *6 gas and liquid chromatography in detail. Understanding Describe the working of electrophoresis machine*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Understanding and densitometer Chromatography And Electrophoresis Chromatographic process - Classification- Terms in chromatography - Gas chromatography: Block diagram - Principle - Constructional details Liquid Chromatography: Types of liquid chromatography- High pressure Liquid Chromatography (HPLC): Principle - Constructional details, Applications in medical field. Electrophoresis apparatus - basic principle and block diagram. Densitometer - principle and block diagram. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding and densitometer Chromatography And Electrophoresis Chromatographic process - Classification- Terms in chromatography - Gas chromatography: Block diagram - Principle - Constructional details Liquid Chromatography: Types of liquid chromatography- High pressure Liquid Chromatography (HPLC): Principle - Constructional details, Applications in medical field. Electrophoresis apparatus - basic principle and block diagram. Densitometer - principle and block diagram*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain 3 Understanding meter. Detail the working of blood cell counters and. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding meter. Detail the working of blood cell counters and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 6 Understanding Hb meter. Describe the principle & construction of. (3 marks)

Answer: Begin with the standard definition or identity of *6 Understanding Hb meter. Describe the principle & construction of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Understanding Blood Gas Analyzer and glucometer pH METER, BLOOD CELL COUNTERS & BLOOD GAS ANALYZER pH meter - Principle of pH measurement - Electrodes for pH measurement - Block diagram of pH meter. Types of blood cells, Blood cell counters - types - coulter counter and flow cytometry. Auto analyzers - block diagram and working. Hb meter - block diagram Blood Gas Analyzer - pO2 and pCO2 measurement - Principle, construction and working of PO2 electrode and PCO2 electrode - Blood Gas Analyzer - block diagram of blood gas analyser. Glucometer - working principle and basic block diagram Text / Reference: T/R Book Title/Author R.S. Khandpur, ‘Handbook of Analytical Instruments’, Tata McGraw Hill T1 publishing Co. Ltd., 2003. T2 G.W. Ewing, Instrumental Methods of Analysis, Mc Graw Hill, 2004. R1 Robrt D Braun’ , ‘Introduction to Instrumental Analysis’ , Mc Graw Hill Ltd Galen W Ewing, ‘Instrumental methods of chemical analysis’ , Mc Graw Hill R2 Ltd Online Resources: Sl.No Website Link 1 https://www.hemakumar.in/analytical-instrumentation 2 https://www.slideshare.net/ShreyaAhuja2/ph-meter-54084084 3 https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html#. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding Blood Gas Analyzer and glucometer pH METER, BLOOD CELL COUNTERS & BLOOD GAS ANALYZER pH meter - Principle of pH measurement - Electrodes for pH measurement - Block diagram of pH meter. Types of blood cells, Blood cell counters - types - coulter counter and flow cytometry. Auto analyzers - block diagram and working. Hb meter - block diagram Blood Gas Analyzer - pO2 and pCO2 measurement - Principle, construction and working of PO2 electrode and PCO2 electrode - Blood Gas Analyzer - block diagram of blood gas analyser. Glucometer - working principle and basic block diagram Text / Reference: T/R Book Title/Author R.S. Khandpur, ‘Handbook of Analytical Instruments’, Tata McGraw Hill T1 publishing Co. Ltd., 2003. T2 G.W. Ewing, Instrumental Methods of Analysis, Mc Graw Hill, 2004. R1 Robrt D Braun’ , ‘Introduction to Instrumental Analysis’ , Mc Graw Hill Ltd Galen W Ewing, ‘Instrumental methods of chemical analysis’ , Mc Graw Hill R2 Ltd Online Resources: Sl.No Website Link 1 https://www.hemakumar.in/analytical-instrumentation 2 https://www.slideshare.net/ShreyaAhuja2/ph-meter-54084084 3 https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html#*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **7 and its performance requirement Summarize the Instrument Calibration**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **9 Understanding and IR spectrophotometers Illustrate the working principle and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain chromatography and its types. 3 Understanding Explain the principle and construction details of**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **3 Understanding meter. Detail the working of blood cell counters and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- https://www.hemakumar.in/analytical-instrumentation
- https://www.slideshare.net/ShreyaAhuja2/ph-meter-54084084
- https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html#

IN-PAGE SEARCH

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